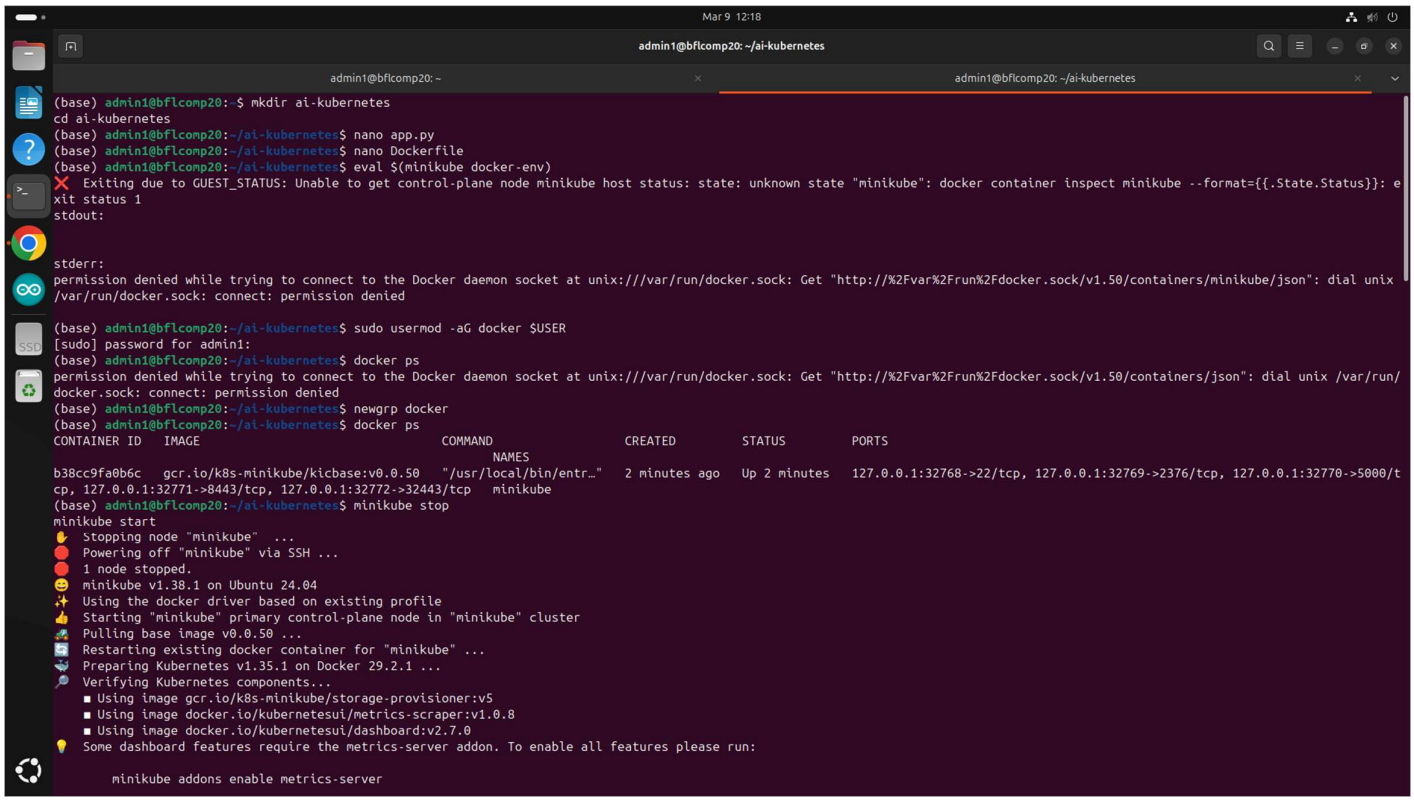
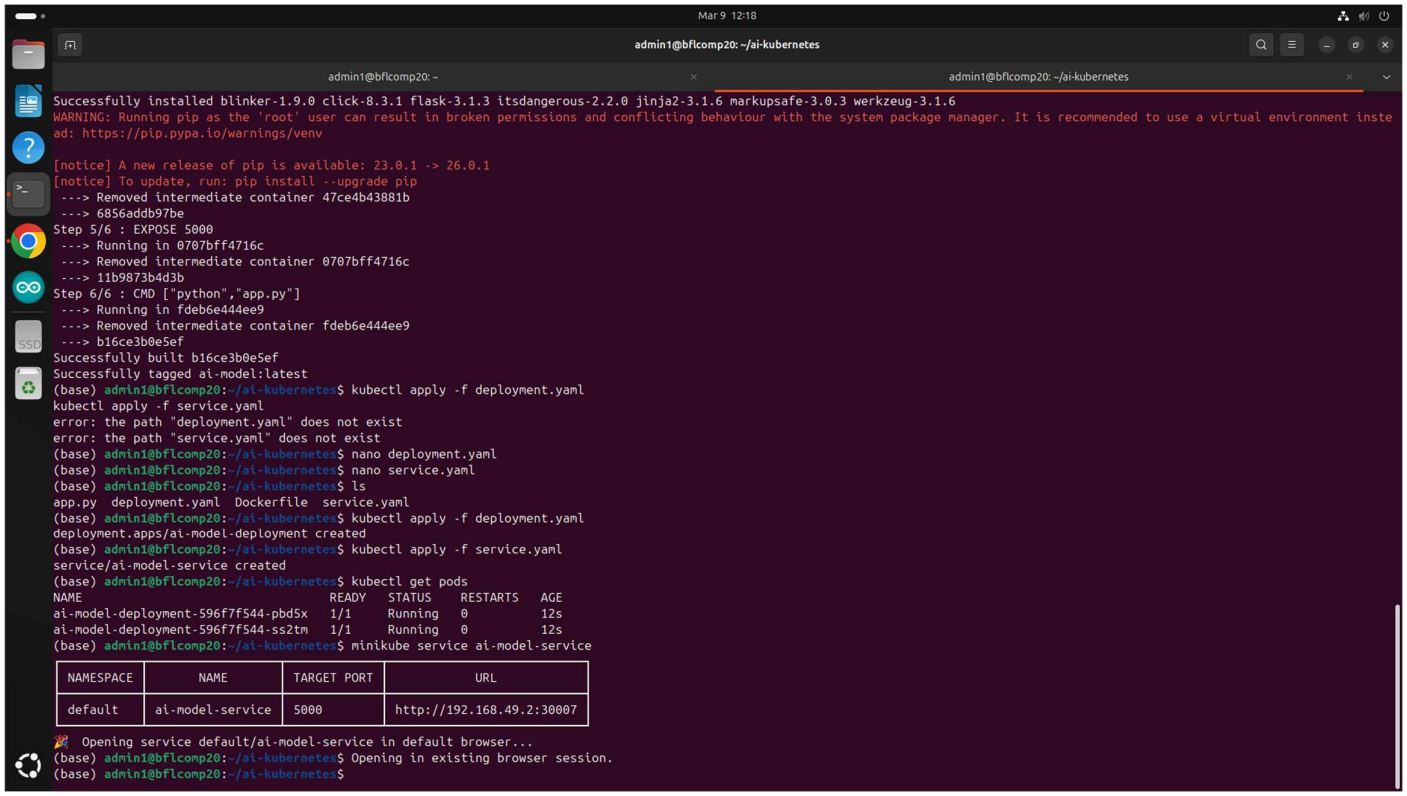
Assignment No.8

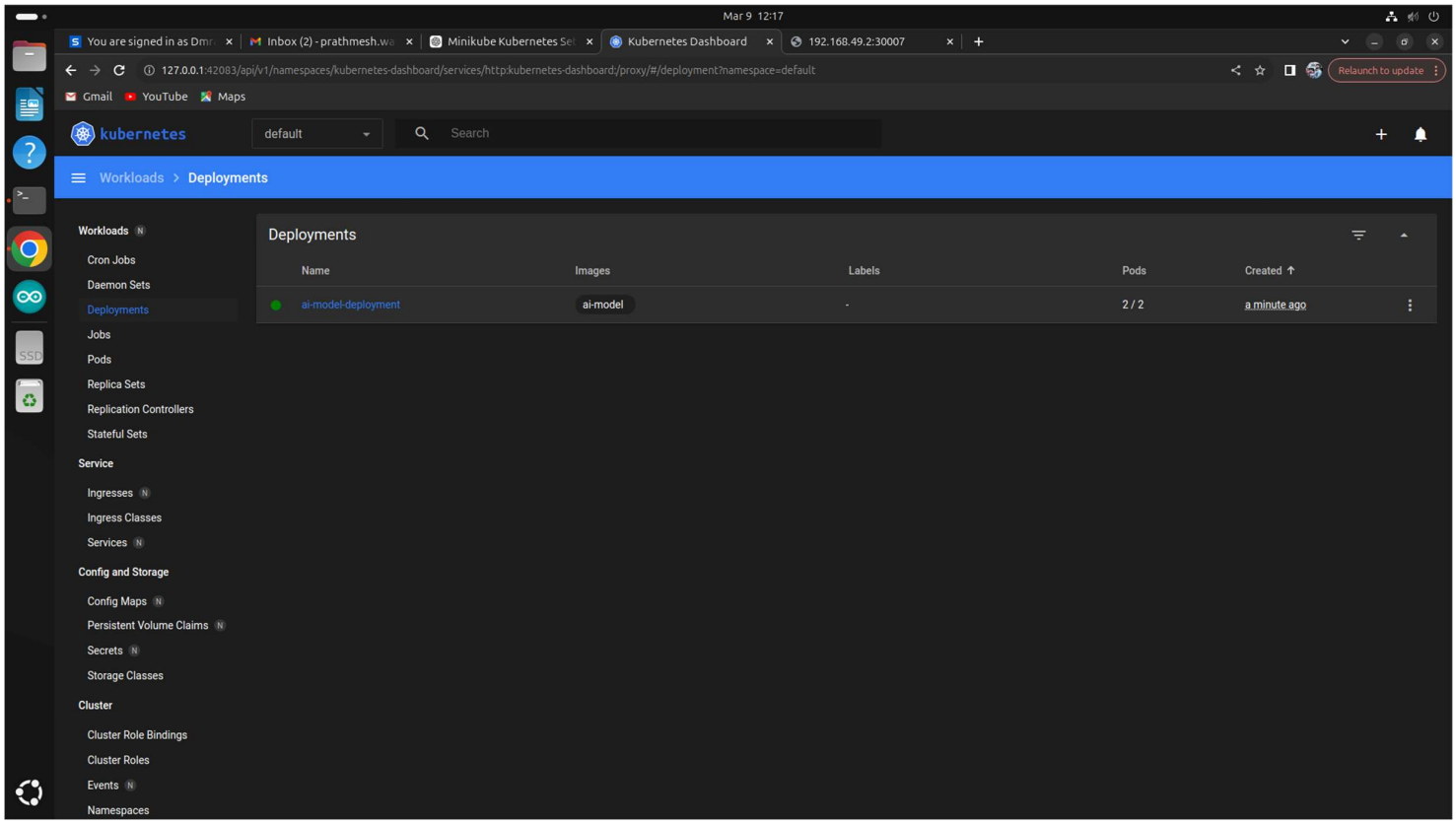
Title: Set up a local Kubernetes cluster with Minikube to deploy and scale your Dockerized AI application.

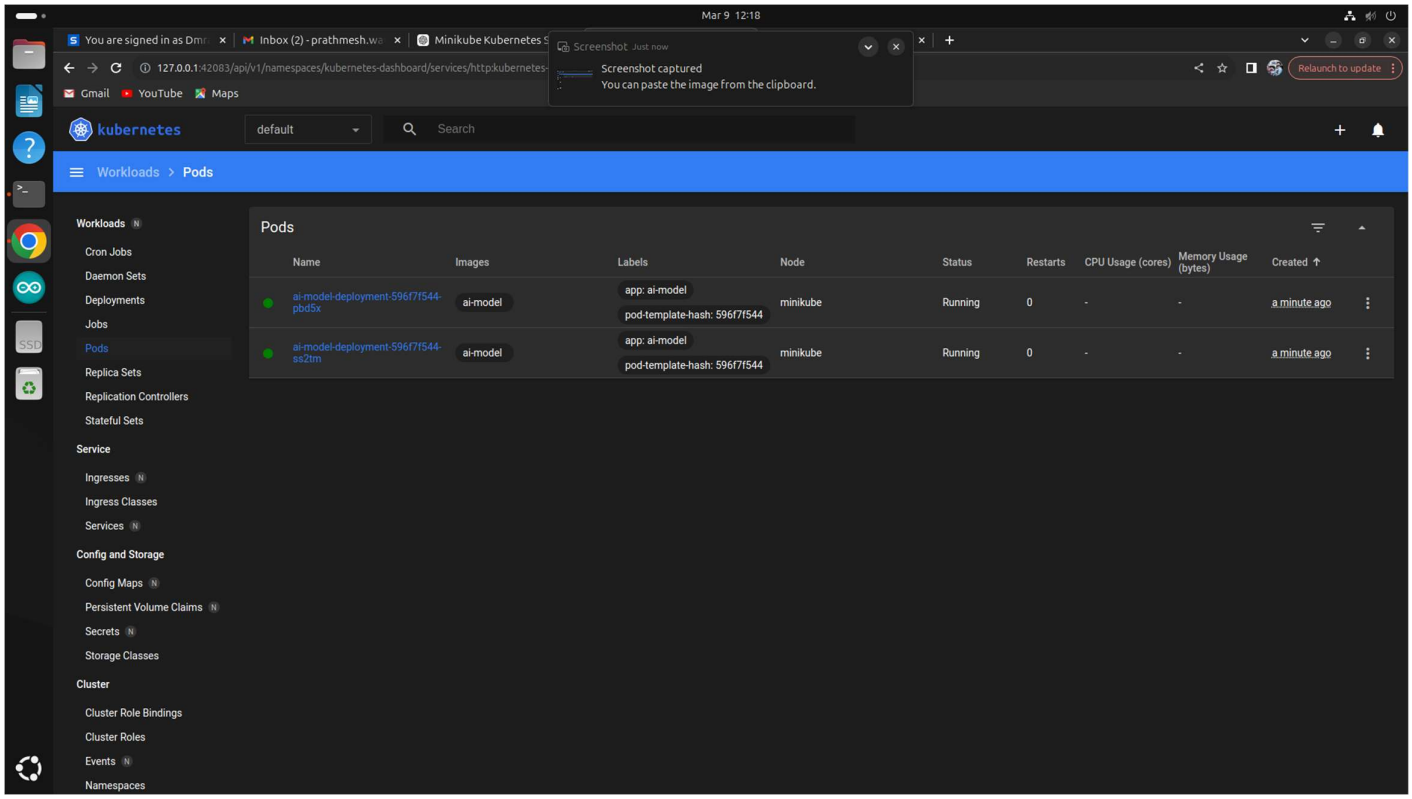
Name: Wavhal Prathmesh Navnath **Div**: TY-B **Roll No:**23107137

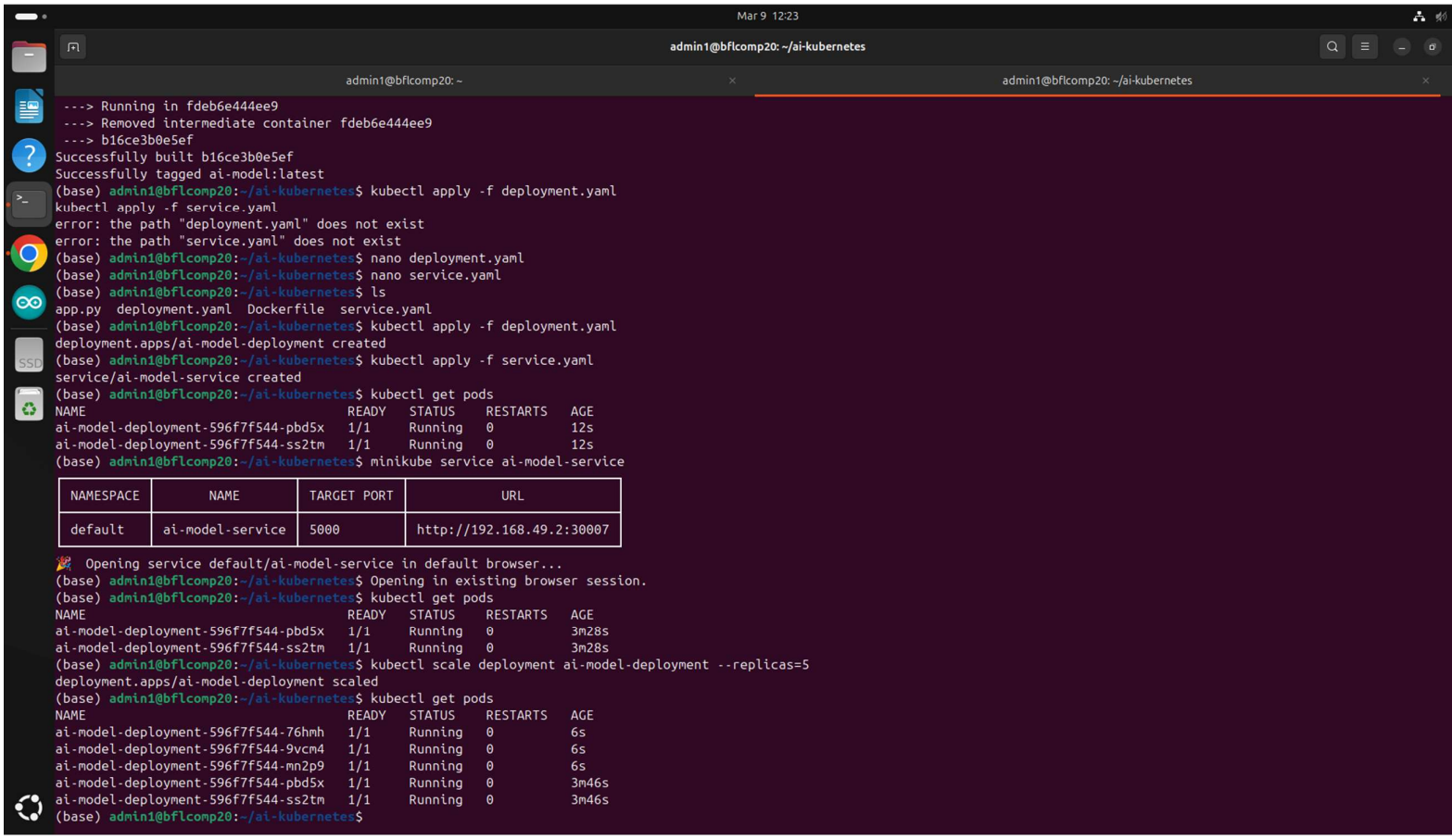
Output:

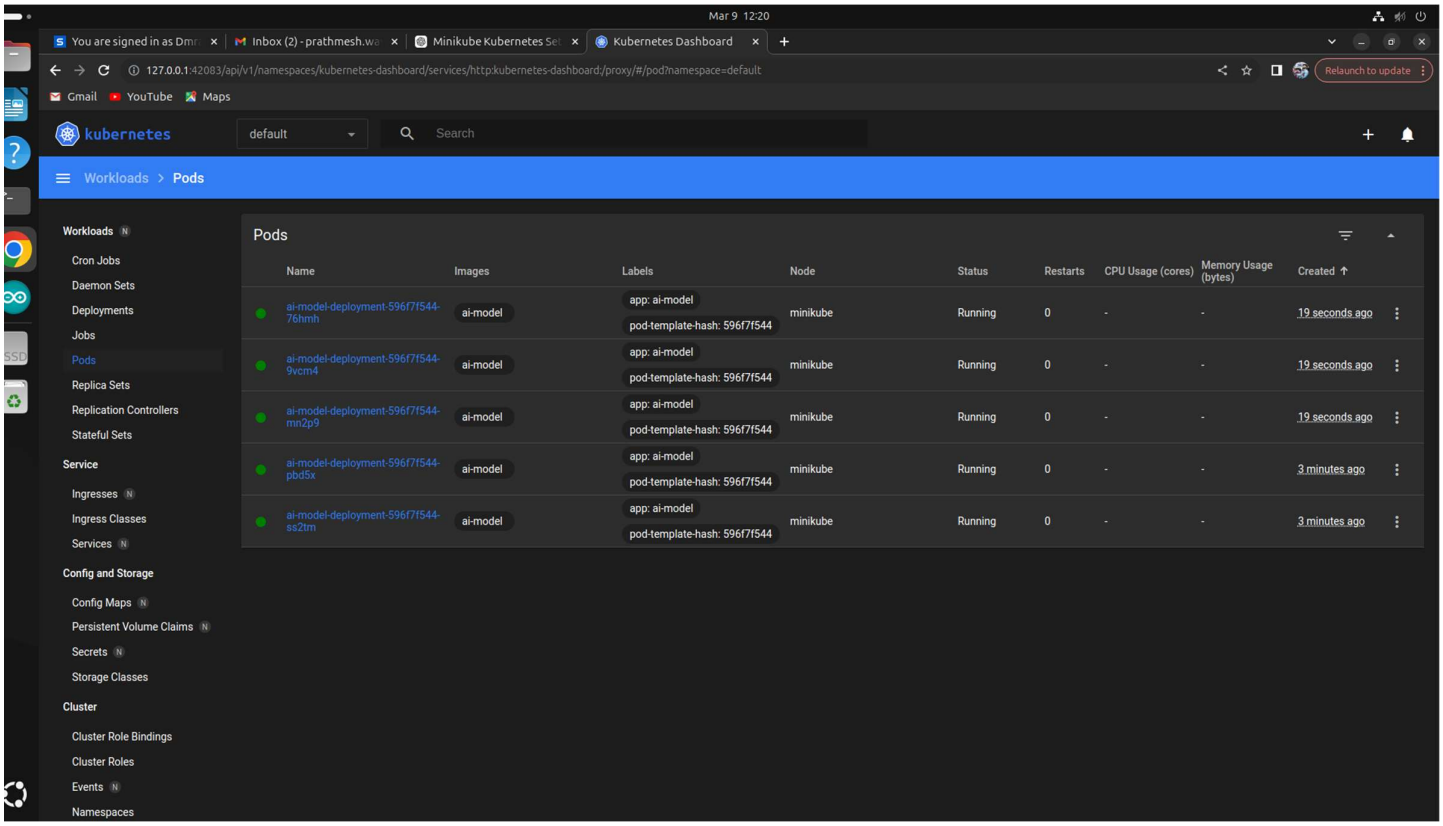


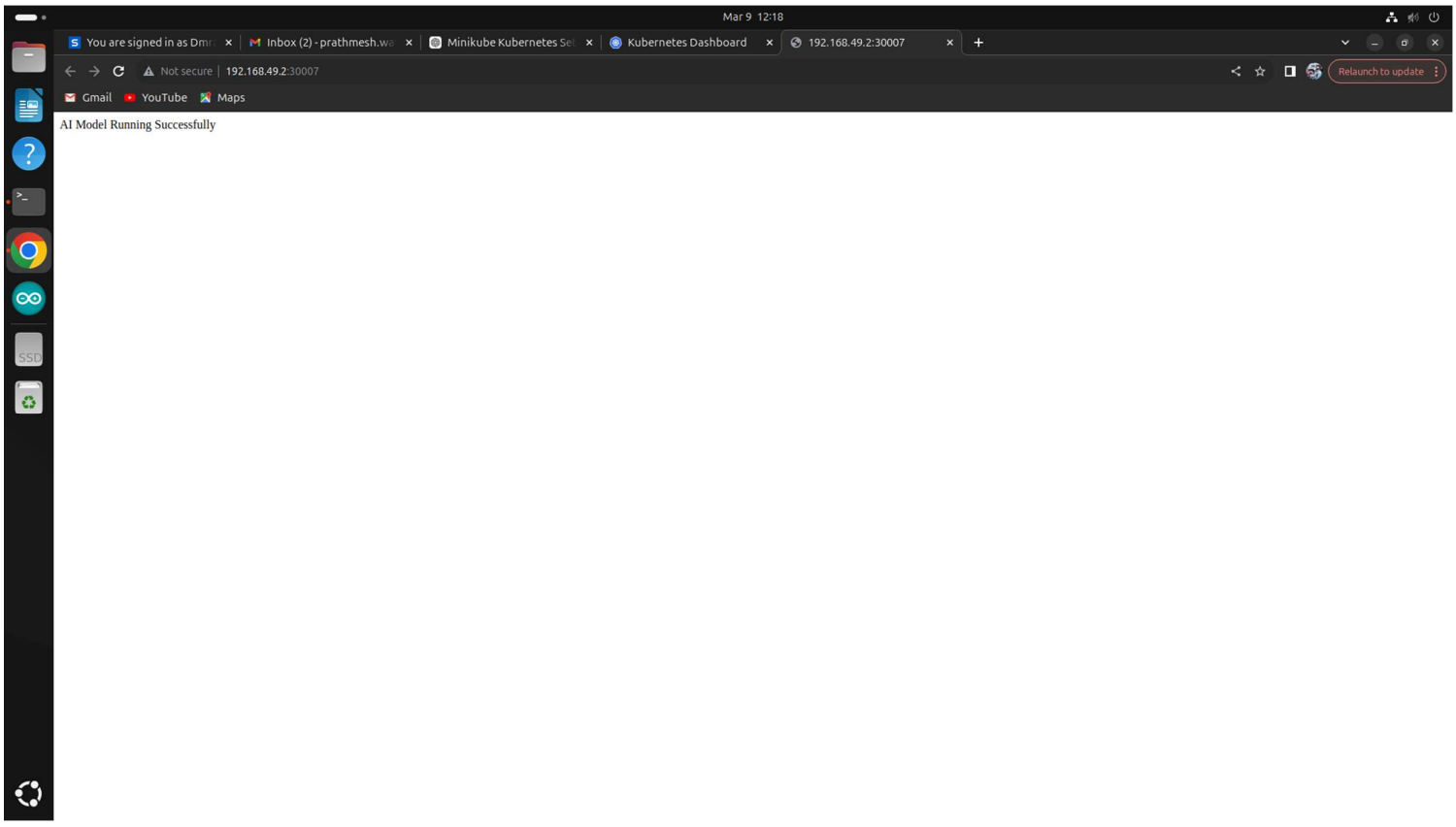

Mar 9 12:18
admin1@bflcomp20: ~/ai-kubernetes
admin1@bflcomp20: ~
admin1@bflcomp20: ~/ai-kubernetes
Successfully installed blinker-1.9.0 click-8.3.1 flask-3.1.3 itsdangerous-2.2.0 jinja2-3.1.6 markupsafe-3.0.3 werkzeug-3.1.6
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv
[notice] A new release of pip is available: 23.0.1 -> 26.0.1
[notice] To update, run: pip install --upgrade pip
---> Removed intermediate container 47ce4b43881b
---> 6856addb97be
Step 5/6 : EXPOSE 5000
---> Running in 0707bff4716c
---> Removed intermediate container 0707bff4716c
---> 11b9873b4d3b
Step 6/6 : CMD ["python","app.py"]
---> Running in fdeb6e444ee9
---> Removed intermediate container fdeb6e444ee9
---> b16ce3b0e5ef
Successfully built b16ce3b0e5ef
Successfully tagged ai-model:latest
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
error: the path "deployment.yaml" does not exist
error: the path "service.yaml" does not exist
(base) admin1@bflcomp20:~/ai-kubernetes$ nano deployment.yaml
(base) admin1@bflcomp20:~/ai-kubernetes$ nano service.yaml
(base) admin1@bflcomp20:~/ai-kubernetes$ ls
app.py deployment.yaml Dockerfile service.yaml
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl apply -f deployment.yaml
deployment.apps/ai-model-deployment created
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl apply -f service.yaml
service/ai-model-service created
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl get pods
NAME READY STATUS RESTARTS AGE
ai-model-deployment-596f7f544-pbd5x 1/1 Running 0 12s
ai-model-deployment-596f7f544-ss2tm 1/1 Running 0 12s
(base) admin1@bflcomp20:~/ai-kubernetes$ minikube service ai-model-service
NAMESPACE NAME TARGET PORT URL
default ai-model-service 5000 http://192.168.49.2:30007
Opening service default/ai-model-service in default browser...
(base) admin1@bflcomp20:~/ai-kubernetes$ Opening in existing browser session.
(base) admin1@bflcomp20:~/ai-kubernetes$


Mar 9 12:17
Kubernetes Dashboard
192.168.49.2:30007
127.0.0.1:42083/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/#/deployment?namespace=default
Relaunch to update
Gmail
YouTube
Maps
kubernetes
default
Search
Workloads > Deployments
Workloads
Cron Jobs
Daemon Sets
Deployments
Jobs
Pods
Replica Sets
Replication Controllers
Stateful Sets
Service
Ingresses
Ingress Classes
Services
Config and Storage
Config Maps
Persistent Volume Claims
Secrets
Storage Classes
Cluster
Cluster Role Bindings
Cluster Roles
Events
Namespaces
Deployments
Name
Images
Labels
Pods
Created
ai-model-deployment
ai-model
-
2 / 2
a minute ago


Mar 9 12:18
Screenshot Just now
Screenshot captured
You can paste the image from the clipboard.
kubernetes
default
Search
Workloads > Pods
Workloads
Cron Jobs
Daemon Sets
Deployments
Jobs
Pods
Replica Sets
Replication Controllers
Stateful Sets
Service
Ingresses
Ingress Classes
Services
Config and Storage
Config Maps
Persistent Volume Claims
Secrets
Storage Classes
Cluster
Cluster Role Bindings
Cluster Roles
Events
Namespaces
Pods
Name
Images
Labels
Node
Status
Restarts
CPU Usage (cores)
Memory Usage (bytes)
Created
ai-model-deployment-596f7f544-pbd5x
ai-model
app: ai-model
pod-template-hash: 596f7f544
minikube
Running
0
a minute ago
ai-model-deployment-596f7f544-ss2tm
ai-model
app: ai-model
pod-template-hash: 596f7f544
minikube
Running
0
a minute ago


Mar 9 12:23
admin1@bflcomp20: ~/ai-kubernetes
admin1@bflcomp20: ~
admin1@bflcomp20: ~/ai-kubernetes
---> Running in fdeb6e444ee9
---> Removed intermediate container fdeb6e444ee9
---> b16ce3b0e5ef
Successfully built b16ce3b0e5ef
Successfully tagged ai-model:latest
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
error: the path "deployment.yaml" does not exist
error: the path "service.yaml" does not exist
(base) admin1@bflcomp20:~/ai-kubernetes$ nano deployment.yaml
(base) admin1@bflcomp20:~/ai-kubernetes$ nano service.yaml
(base) admin1@bflcomp20:~/ai-kubernetes$ ls
app.py deployment.yaml Dockerfile service.yaml
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl apply -f deployment.yaml
deployment.apps/ai-model-deployment created
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl apply -f service.yaml
service/ai-model-service created
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl get pods
NAME READY STATUS RESTARTS AGE
ai-model-deployment-596f7f544-pbd5x 1/1 Running 0 12s
ai-model-deployment-596f7f544-ss2tm 1/1 Running 0 12s
(base) admin1@bflcomp20:~/ai-kubernetes$ minikube service ai-model-service
NAMESPACE NAME TARGET PORT URL
default ai-model-service 5000 http://192.168.49.2:30007
Opening service default/ai-model-service in default browser...
(base) admin1@bflcomp20:~/ai-kubernetes$ Opening in existing browser session.
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl get pods
NAME READY STATUS RESTARTS AGE
ai-model-deployment-596f7f544-pbd5x 1/1 Running 0 3m28s
ai-model-deployment-596f7f544-ss2tm 1/1 Running 0 3m28s
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl scale deployment ai-model-deployment --replicas=5
deployment.apps/ai-model-deployment scaled
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl get pods
NAME READY STATUS RESTARTS AGE
ai-model-deployment-596f7f544-76hmh 1/1 Running 0 6s
ai-model-deployment-596f7f544-9vcm4 1/1 Running 0 6s
ai-model-deployment-596f7f544-mn2p9 1/1 Running 0 6s
ai-model-deployment-596f7f544-pbd5x 1/1 Running 0 3m46s
ai-model-deployment-596f7f544-ss2tm 1/1 Running 0 3m46s
(base) admin1@bflcomp20:~/ai-kubernetes$


Mar 9 12:20
Kubernetes Dashboard
127.0.0.1:42083/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/#/pod?namespace=default
Relaunch to update
kubernetes
default
Search
Workloads > Pods
Workloads
Cron Jobs
Daemon Sets
Deployments
Jobs
Pods
Replica Sets
Replication Controllers
Stateful Sets
Service
Ingresses
Ingress Classes
Services
Config and Storage
Config Maps
Persistent Volume Claims
Secrets
Storage Classes
Cluster
Cluster Role Bindings
Cluster Roles
Events
Namespaces
Pods
Name
Images
Labels
Node
Status
Restarts
CPU Usage (cores)
Memory Usage (bytes)
Created
ai-model-deployment-596f7f544-76hmh
ai-model-deployment-596f7f544-9vcm4
ai-model-deployment-596f7f544-mn2p9
ai-model-deployment-596f7f544-pbd5x
ai-model-deployment-596f7f544-ss2tm
ai-model
app: ai-model
pod-template-hash: 596f7f544
minikube
Running
0
19 seconds ago
3 minutes ago


Mar 9 12:18
Minikube Kubernetes Set
Kubernetes Dashboard
192.168.49.2:30007
Not secure | 192.168.49.2:30007
Relaunch to update
Gmail
YouTube
Maps
AI Model Running Successfully